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Design and Simulation of a Broadband Tunable Transmittance Optical Filter Using Sb₂Se₃ Phase Change Material

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ABSTRACT We propose a switchable transmittance filter that operates around 1064 nm wavelengths using antimony triselenide (Sb₂Se₃). Sb₂Se₃ is a phase change material (PCM) that has low absorption in the infrared (IR) spectrum, large and stable refractive index contrast between the two phases, and high optical endurance of $\sim 10^6$ cycles. We provide the reasoning for selecting a long-pass edge-filter design and compare its performance to other planar thin-film designs. At 1064-nm, the selected 5-unit-cell edge-filter design has 93.6% transmittance in amorphous phase and 0.2% in crystalline phase, a 93.4% transmittance contrast with ~ 200 nm FWHM. It has minimal transmittance degradation after enduring over 1.5×10^6 phase switching cycles. We show the transmittance tunability of the design by changing the phase of each Sb₂Se₃ layer, and each layer thickness can be optimized to increase the optical performance across the 900-1200 nm wavelengths.

INDEX TERMS Antimony triselenide, Optical designs, Optical filters, Phase change materials, Thin films

I. INTRODUCTION

The term phase change material (PCM) can have different meanings depending on the research field. In energy storage, the term refers to materials that can store and release its thermal energy though changes in its state of matter, e.g. from solid to liquid state [1]. However, in the fields of electronics and photonics, the term refers to a solid-state material that changes its crystal structure when depending on the temperature, such as from amorphous to crystalline phase when temperature rise above transition temperature (T_{trans}). This change leads to a high contrast in complex refractive index $\tilde{n}$ between the initial and post-transitioned phases, where $\tilde{n} = n + ik$ comprises of the real refractive index n and the imaginary extinction coefficient k [2]. The refractive index contrast is an order of magnitude larger than the contrast from applied electric field or Kerr effect (light intensity) [3], this property makes PCMs ideal for photonic applications that require the modulation of optical performance. Some examples are reconfigurable metasurfaces [2, 4-6], optical

limiters [3, 7-11], optical switches [12, 13], optical filters [14-17], and absorbers [18, 19].

The classification of PCM can be done by its behaviour post phase transition. It is considered a volatile PCM if the phase reverts back to pre-transition phase when the temperature falls below T_{trans} . It is considered a non-volatile PCM if it maintains its post-transition phase when temperature falls below T_{trans} , a way to changes the phase back to pre-transition is to melt and then rapidly quench to a temperature below T_{trans} . The quenching speed needs to be fast enough such that the atoms do not have time to form orderly bonds. One of the most common volatile PCM is vanadium dioxide (VO₂), it undergoes metal-to-insulation transition (MIT) at 68 °C, the lowest among any PCMs. VO₂ has a relatively large complex refractive index contrast (Δn) in the near-IR and IR spectra [20], but its biggest downside is the large absorption due to high extinction coefficients in both phases, this limits the maximum thickness a design can have before the absorption is too high for efficient devices. A common non-volatile

PCMs are chalcogenide glasses called $\text{Ge}_2\text{Sb}_2\text{Te}_5$ (GST) and $\text{Ge}_2\text{Sb}_2\text{Se}_4\text{Te}$ (GSST). GST beings one of the earliest and most studied PCM [21], it has large refractive index contrast between amorphous and crystalline phases, relative low T_{trans} at $\sim 145^\circ\text{C}$ when compared to other PCMs [3, 22]. Meanwhile GSST, which is a derivative of GST that substitutes parts of tellurium (Te) with selenium (Se), has lower extinction coefficients than either GST or VO_2 . But when compared to GST, it has higher T_{trans} at $\sim 242^\circ\text{C}$, and the Δn is also lower [22].

With that being said, antimony triselenide (Sb_2Se_3) is a non-volatile PCM which originated from solar-energy applications, but it has very few research in the field of photonics. Sb_2Se_3 unique properties are it near-zero extinction coefficients in near IR and IR spectra, its high and stable refractive index contrast, and very high phase switching endurance in an order of 10^6 – 10^7 cycles [23, 24]. Its transition temperature is in between GST and GSST, at $\sim 186^\circ\text{C}$ [23, 25]. These properties make Sb_2Se_3 an attractive PCM for low-loss, reconfigurable devices used as optical switches or programmable circuits in optical waveguides [4–6, 13, 15, 21]. But its usage as a tunable transmittance filters has not been explored much [16].

The transmittance of an optical filter can be modulated by the changes in refractive index of PCM. The ON mode has high transmission by creating antireflection condition and minimizing the absorption from the filter, and OFF mode has low transmission after phase transition. The method for switching Sb_2Se_3 phase could be from laser-induced heating [23], or from electrical heating [14]. The switching via laser heating cannot be from infrared light since Sb_2Se_3 has very small absorption at that wavelengths. The architecture can be multilayer planar thin-film structures or nanostructure metasurfaces; planar structures are more straightforward to design and fabricate, while metasurfaces offer higher design flexibility [10]. For planar designs, there are many methods for creating antireflection condition, such as single-layer or multilayer antireflection designs, line-pass filters, or Fabry–Pérot resonant cavities [17, 26].

In this work, we investigate Sb_2Se_3 as a PCM in planar thin-film designs for transmittance tunable optical filters. The operating wavelength is selected to be 1064 nm, Nd:YAG laser wavelength which is commonly used in laboratory setups. At this wavelength, Sb_2Se_3 exhibits zero absorption in the amorphous phase and near-zero absorption in the crystalline phase, and its refractive index contrast is high and stable throughout the IR spectrum. The goal is to create a low-loss, broadband transmittance-tunable optical filter.

II. THEORIES

A. OPTICAL PROPERTIES OF ANTIMONY TRISELENIDE

Antimony triselenide (Sb_2Se_3) is an orthorhombic chalcogenide non-volatile phase change material (PCM) characterized by quasi-one-dimensional ribbons. While its amorphous phase consists of a disordered network of

distorted motifs, crystallization restores long-range order along the ribbon direction. This thermally driven transition can be triggered by nanosecond to microsecond optical or electrical pulses [21, 23]. The process is reversible and has high endurance, with optical switching reaching 10^6 to 10^7 cycles [23, 24]. As an optical device, Sb_2Se_3 enables repeatable toggling between high and low transmission modes, provided that local heating remains below the melting or ablation threshold [4, 24].

The complex refractive index $\tilde{n}$ of any phase change material at wavelength λ is:

$$\tilde{n}_p(\lambda) = n_p(\lambda) + ik_p(\lambda) \quad (1)$$

where p denotes the phase of Sb_2Se_3 which can be either amorphous (*amo*) or crystalline (*cry*) phase. Dispersive models, such as Tauc-Lorentz or multi-oscillator fits are often used to represent the measured n_p and k_p over the spectral range of interest. However, in this work, linear interpolation is used because all the datasets used have measurement interval of less than 4 nm. Figure 1 shows their complex refractive indices from 300 to 1700 nm using the dataset from Delaney, et al. [21], where the complex refractive indices at 1064 nm are $\tilde{n}_{amo} = 3.370 + 0.000i$ and $\tilde{n}_{cry} = 4.308 + 0.010i$. The blue lines represent real refractive index (n) and the red lines represent imaginary extinction coefficient (k). In amorphous phase, the refractive index increases from ~ 2.4 at 300 nm to a peak of ~ 4.0 at 600 nm and gradually decreases to a stable value of ~ 3.3 above 1200 nm, indicating a strong dispersion near the band edge of visible spectrum and a weaker dispersion in the NIR spectrum. The extinction coefficient starts from ~ 2.1 at 300 nm and drops to zero at ~ 850 nm. In the crystalline phase, the refractive index at the UV wavelengths is around the same value as the amorphous phase but reaches a higher peak of ~ 5.1 at around 660 nm; it then drops to a stable value of ~ 4.1 in the NIR spectrum. The absorption index of crystalline phase peaks at ~ 3.1 in the near-UV spectrum and rapidly approaches zero above 900 nm.

The refractive index contrasts ($\Delta n = n_c - n_a$) are large and stable from the visible through IR spectra, and the extinction coefficients are large in the near-UV region but fall to zero around the NIR spectrum for both phases, with the crystalline phase exhibiting higher absorption in the visible spectrum. Thus, when compared to other PCMs, Sb_2Se_3 has several advantages when operating in the NIR and IR wavelengths. The large Δn and low k starting from NIR wavelengths enable a device to operate in two modes with little to no optical loss, and a stable Δn also helps with broadband designs.

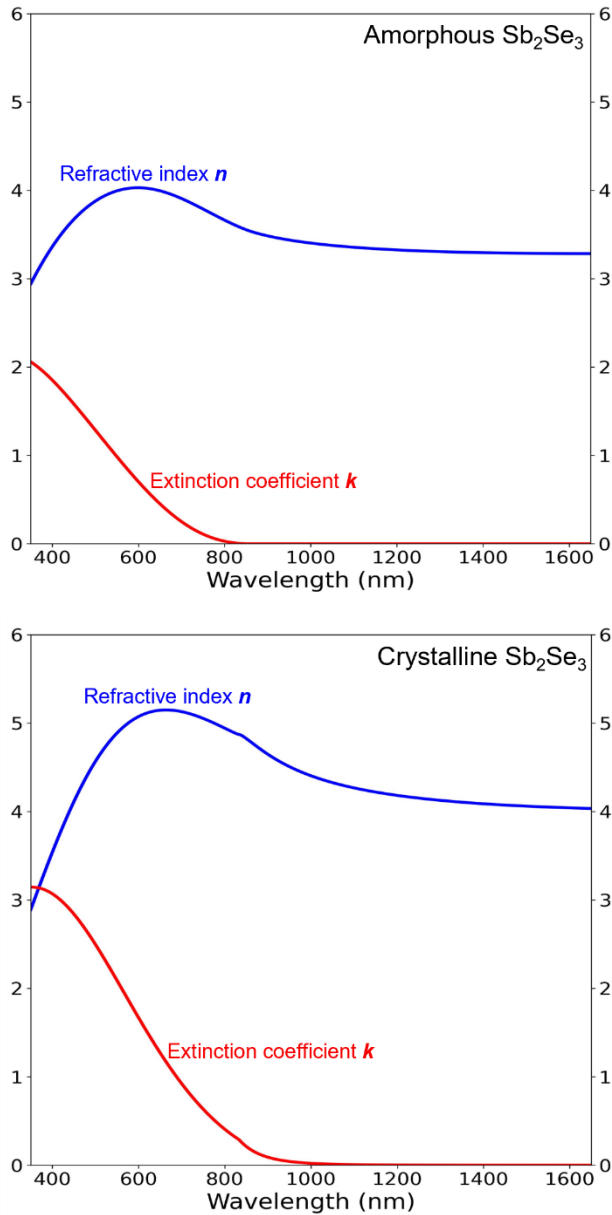


FIGURE 1. Complex refractive index of Sb_2Se_3 . Graphs are plotted from experimental data of [21].

B. SINGLE-LAYER ANTIREFLECTION DESIGN

The goal of designing most optical devices is to either create antireflection or high-reflection conditions. For a two-mode optical device, such as a transmittance tunable filter, we can consider the reflection in the ON mode first, and then the reflectance contrast ΔR between the ON and OFF modes. The simplest antireflection from a planar thin-film design is one film layer deposited on top of a substrate; both the film and the substrate should have zero absorption. The diagram in the top right corner of Fig. 2 shows that a single-layer film reflects light at two interfaces: the bottom interface between the film and substrate (R_1) and the top interface between the air and film (R_2). Antireflection occurs from the destructive interference between R_1 and R_2 ,

with a complete AR condition requiring the same amplitude and a phase difference of π radian (optical path length of $\lambda/2$). Because R_1 propagates through the thin film layer twice (while transmitted down and reflected backup), the optical thickness (OT) of the film only needs to be equal to half of the optical path length, $OT = \lambda/4$, this also known as quarter-wave optical thickness (QWOT). The optical thickness is related to the physical thickness t by the following equation:

$$OT(\lambda) = n(\lambda) t, \quad (2)$$

where $n(\lambda)$ is the refractive index of the film at λ . For a single-layer AR device, the thin-film physical thickness t must be

$$t = \frac{OT}{n} = \frac{\lambda/4}{n}. \quad (3)$$

Another method for calculating the thickness required for the AR condition is to consider the reflection of the device. The reflectance R of only the substrate is related to the Fresnel coefficient r by

$$R = |r|^2 = \left| \frac{n_a - n_s}{n_a + n_s} \right|^2, \quad (4)$$

where n_a is the air refractive index, and n_s is the refractive index of substrate. If a single-layer film is placed on top of a substrate, the film and substrate layers can be combined into an effective reflectance index n_r [26], defined as

$$n_r = n_f \frac{(n_s + n_f) + (n_s - n_f)e^{-i2\theta}}{(n_s + n_f) - (n_s - n_f)e^{-i2\theta}}, \quad (5)$$

where n_f is the film refractive index, i is the imaginary unit, and θ is the phase thickness which is related to optical thickness and physical thickness t by the angular wavenumber $k_0 = 2\pi/\lambda$:

$$\theta = k_0 OT = k_0 nt. \quad (6)$$

By replacing n_s in Eq. 4 with n_r from Eq. 5, the reflectance of a single-layer device becomes:

$$R = |r|^2 = \left| \frac{n_a - n_r}{n_a + n_r} \right|^2. \quad (7)$$

The antireflection condition happens when $R = 0$ or $n_r = n_a$. This condition occurs when the phase thickness θ is in multiple of $\pi/2$ radians:

$$\theta = Z \frac{\pi}{2}, \quad (8)$$

where Z denotes even integer. If this condition is satisfied, Eq. 5 can be simplified with Euler's identity ($\exp^{-i\pi} = -1$) to:

$$n_r = n_f \frac{(n_s + n_f) - (n_s - n_f)}{(n_s + n_f) + (n_s - n_f)} = \frac{n_f^2}{n_s}. \quad (9)$$

By substituting this into Eq. 7, the refractive indices for antireflection condition becomes:

$$n_a n_s = n_f^2 \quad (10)$$

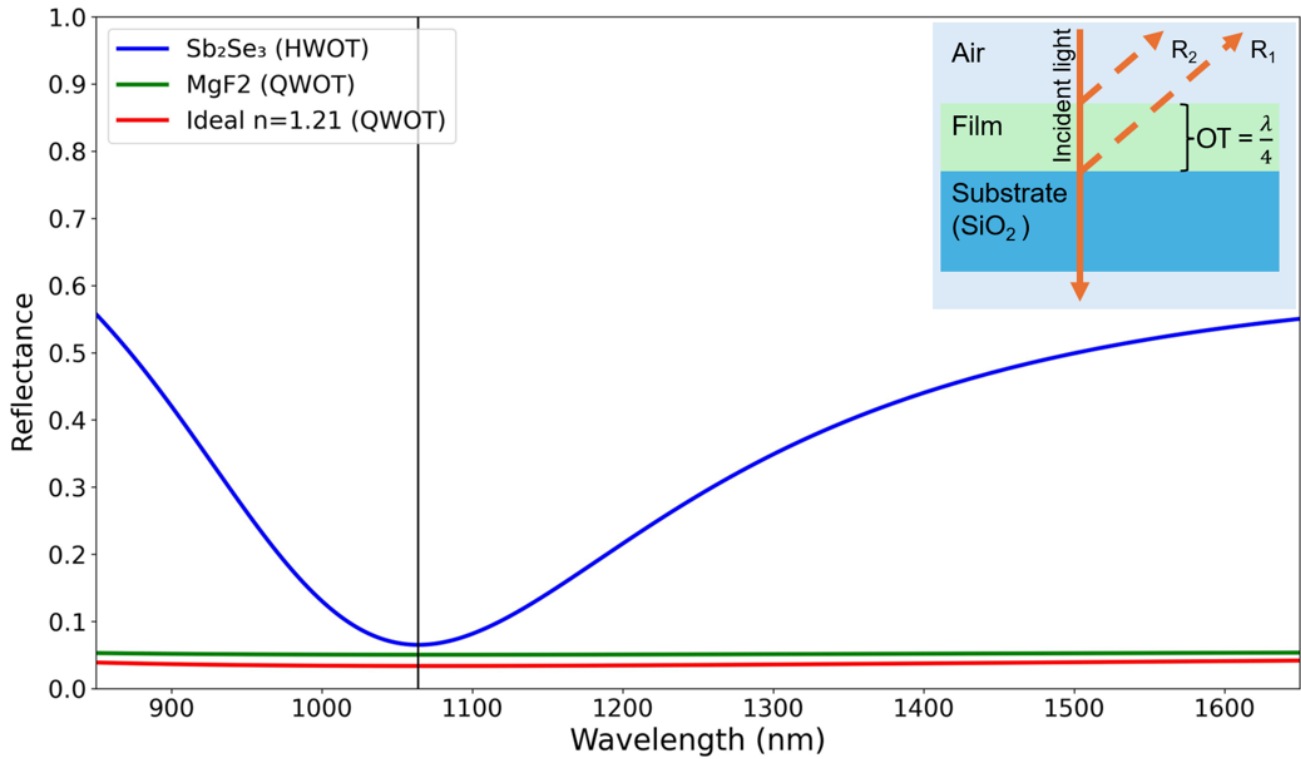


FIGURE 2. The reflectance of a single-layer design with different refractive index and thickness, the black vertical line indicates 1064 nm. At this wavelength, HWOT-thick Sb_2Se_3 (blue) has the highest reflection $R = 6.5\%$. QWOT-thick MgF_2 (green) $R = 5.0\%$. An ideal QWOT-thick film with $n_{\text{ideal}} = 1.21$ (red) has the lowest reflectance, $R = 3.4\%$. Cross-section diagram of the QWOT-thick structure is shown in the top-right corner.

Another condition is the relative refractive index between the air, film, and substrate, which must be in the following order: $n_{\text{air}} < n_f < n_{\text{substrate}}$. Light shifts its phase by π radian when reflected from a lower to a higher refractive index medium. The reflected light at the air-film interface (R_2) will always undergo a π radian phase shift since n_a will be the lowest in most cases. Therefore, R_1 must undergo the same phase shift to create destructive interference.

These two conditions limit material selections for a perfect antireflection, since there is only one ideal refractive index (n_{ideal}) for each n_s . The refractive index of Sb_2Se_3 is higher than that of substrate (silica; SiO_2) at every wavelength. So, if Sb_2Se_3 is QWOT, the reflection from R_1 still travels 2QWOT (π radian) further than R_2 but only R_2 has its phase shifted by π radian during reflection, so the two R 's have a phase difference of 0 radian, a constructive interference. To minimize reflection, the Sb_2Se_3 thickness can be increased to twice the QWOT ($OT = \lambda/2$), this is called half-wave optical thickness (HWOT) and is equivalent to an absentee layer. For example, at the reference wavelength $\lambda_0 = 1064$ nm, if we use 3-mm-thick SiO_2 as the substrate ($n_s = 1.46$) [27], the ideal QWOT-film thickness is calculated from Eq.10 to be $n_{\text{ideal}} = 1.21$. The material with the closest refractive index is MgF_2 ($n_{\text{MgF}_2} = 1.38$) [28]. A slight difference from ideal value causes the reflection to be slightly higher, as illustrated in Fig. 2. The reflectance of the Sb_2Se_3 (blue) is the highest, at 6.5%, because the PCM

layer is an absentee layer, so the reflection is equivalent to a standalone 3-mm SiO_2 substrate. An ideal film (red) still exhibits small reflectance because of the reflection in the substrate-air layer in the back.

C. MULTILAYER ANTIREFLECTION DESIGN

Two-layer design with each layer thickness being QWOT (Q/Q design) is also be used to create antireflection conditions. The Fresnel reflection coefficient r from Eq. 7 can be expanded to

$$r = \frac{n_a - n_{r2}}{n_a + n_{r2}} = \frac{n_a - \left(\frac{n_{f2}^2}{n_{r1}}\right)}{n_a + \left(\frac{n_{f2}^2}{n_{r1}}\right)} = \frac{n_a - \left(\frac{n_{f2}^2}{n_{f1}/n_s}\right)}{n_a + \left(\frac{n_{f2}^2}{n_{f1}/n_s}\right)}, \quad (11)$$

where n_f begins from the substrate side and increases toward the air side ($n_s / n_{f1} / n_{f2} / n_a$), n_{r2} is the effective reflectance index of the entire structure [26], and n_{r1} is the effective reflectance index of only the substrate (n_s) and bottom film (n_{f1}). Similar to Eq. 7, the antireflection condition of the two-layer design occurs when $n_a = n_{r2}$ or

$$n_a = \frac{n_{f2}^2}{n_{f1}^2/n_s}, \quad (12)$$

$$\frac{n_a}{n_s} = \frac{n_{f2}^2}{n_{f1}^2}. \quad (13)$$

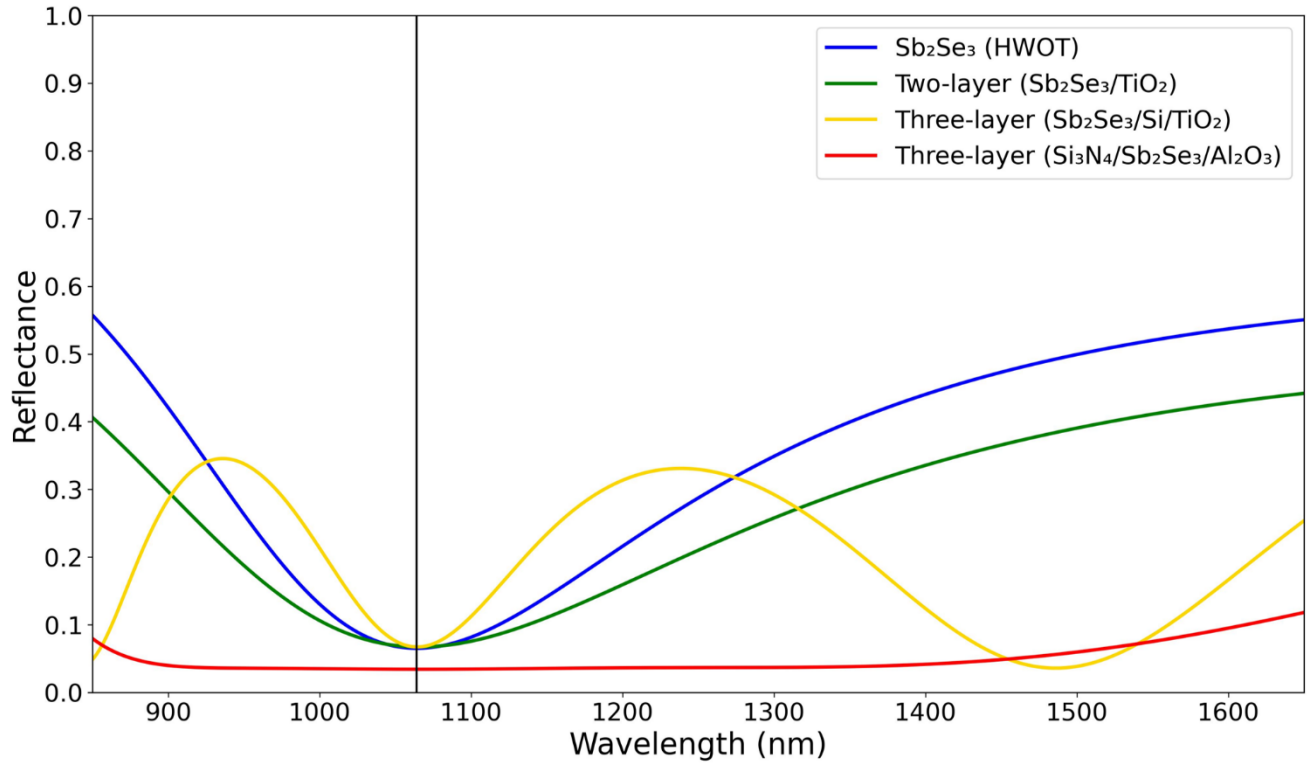


FIGURE 3. The reflectance profiles of four different designs. The single-layer, two-layer, and three-layer design with Sb_2Se_3 at the bottom have around the $R = 6.5\text{--}6.7\%$. However, the three-layer design with Sb_2Se_3 as middle layer has refractive index ratio of n_{f1} and n_{f2} closer to ideal values, so its reflectance value is the lowest at $R = 3.4\%$.

From Eq. 13, it can be seen that the refractive index of the top film must be smaller than the bottom film ($n_{f2} < n_{f1}$). For example, at $\lambda_0 = 1064$ nm, if we use the same SiO_2 substrate ($n_s = 1.46$) and Sb_2Se_3 as the bottom film ($n_{f1} = 3.37$), the ideal refractive index of the top film (n_{f2}) is 2.78. A commonly used material with the closest refractive index is TiO_2 ($n_{\text{TiO}_2} = 2.31$) [29]. Compared to the single-layer Sb_2Se_3 design, the two-layer $\text{Sb}_2\text{Se}_3/\text{TiO}_2$ design has a lower reflectance at 1064 nm and wider AR region, as shown in Fig. 3. Another way to create a two-layer AR condition is to fix the two film refractive indices and vary the thickness of each film, as explained in Chapter 5.2 of [26]. However, this method narrows the spectral range of the antireflection region.

A three-layer design with an HWOT-thick film in the middle (Q/H/Q) requires refractive index orderings of $n_s < n_{f1} < n_{f2}$ and $n_{f2} > n_{f3} > n_a$ for antireflection properties. The middle layer (n_{f2}) acts as an absentee layer when the incident wavelength is the same as the reference wavelength ($\lambda = \lambda_0$). The reflection coefficient is equal to

$$r = \frac{n_a - n_{f3}}{n_a + n_{f3}} = \frac{n_a - \left(\frac{n_{f3}^2}{n_{f2}}\right)}{n_a + \left(\frac{n_{f3}^2}{n_{f2}}\right)}. \quad (14)$$

Because the middle film layer (n_{f2}) is HWOT thick, it behaves as an absentee layer [26, 30]. So $n_{f2} = n_{f1}$, and Eq. 14 becomes:

$$r = \frac{n_a - \left(\frac{n_{f3}^2}{n_{f1}}\right)}{n_a + \left(\frac{n_{f3}^2}{n_{f1}}\right)} = \frac{n_a - \left(\frac{n_{f3}^2}{n_{f1}^2/n_s}\right)}{n_a + \left(\frac{n_{f3}^2}{n_{f1}^2/n_s}\right)}, \quad (15)$$

and a three-layer antireflection condition occurs when

$$\frac{n_a}{n_s} = \frac{n_{f3}^2}{n_{f1}^2}. \quad (16)$$

The refractive index ratio of top and bottom films is similar to that of the two-layer design. As an example, if the bottom layer (n_{f1}) is Sb_2Se_3 (QWOT), then the ideal top layer ($n_{f3, \text{ideal}}$) is 2.78, so TiO_2 ($n_{\text{TiO}_2} = 2.31$) is used again. The middle layer needs to have the highest refractive index, so Silicon (Si) is used ($n_{\text{Si}} = n_{f2} = 3.55$) [31]. As shown in Fig. 3, the reflectance of the three-layer design (yellow) at wavelengths further from λ_0 (i.e., 1400 nm and longer) is lower than that of a single (blue) or two-layer design (green). This is a benefit of adding an extra absentee layer in the middle, it contributes to a broader operating bandwidth [30].

Instead of placing the PCM as the bottom film layer, Sb_2Se_3 can be moved to the middle so that the selection of ideals n_{f1} and n_{f3} is easier. The bottom layer could be Si_3N_4 ($n_{f1} = 2.00$) [32] and the ideal top layer calculated using Eq. 16 has $n_{f3} = 1.65$, which Al_2O_3 has the closest refractive index ($n_{\text{Al}_2\text{O}_3} = 1.62$) [29]. These designs are shown in Fig. 3, where the reflectance of the designs with Sb_2Se_3 deposited directly on

top of the substrate, i.e., single-layer, two-layer, and three-layer designs, are $\sim 6.5\%$. But the three-layer design with Sb_2Se_3 placed in the middle ($n_2 = \text{Sb}_2\text{Se}_3$) has a reflectance of 3.4% , which is the same value as that of an ideal single-layer design (3.4%). This improvement can be explained by the selection of n_1 and n_3 which are closer to ideal values.

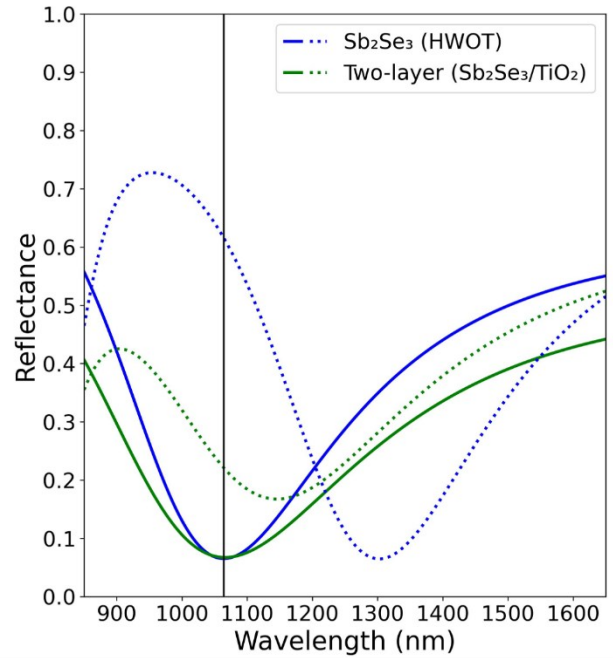
D. LONG-PASS EDGE-FILTER DESIGN

In previous sections, we only consider the reflectance of the design when Sb_2Se_3 is in amorphous phase (ON mode). However, because we want to modulate the performance of our optical filters, the reflectance when Sb_2Se_3 transition to crystalline phase (OFF mode) should also be considered. A high reflectance contrast between crystalline and amorphous phases ($\Delta R = R_c - R_a$) indicates a good modulation power. The reflectance plots of the four designs from Fig. 3 are shown in Fig. 4. The single-layer design reflectance profile red-shifted for ~ 250 nm and is the only one that has reflectance contrast higher than 50% (Fig 4a, blue: $\Delta R=55\%$). More importantly, the three-layer design with Sb_2Se_3 in the middle, which has the lowest reflectance in amorphous phase ($R_a = 3.4\%$), only has $R_c = 14.8\%$ (Fig 4b, red: $\Delta R=11.4\%$).

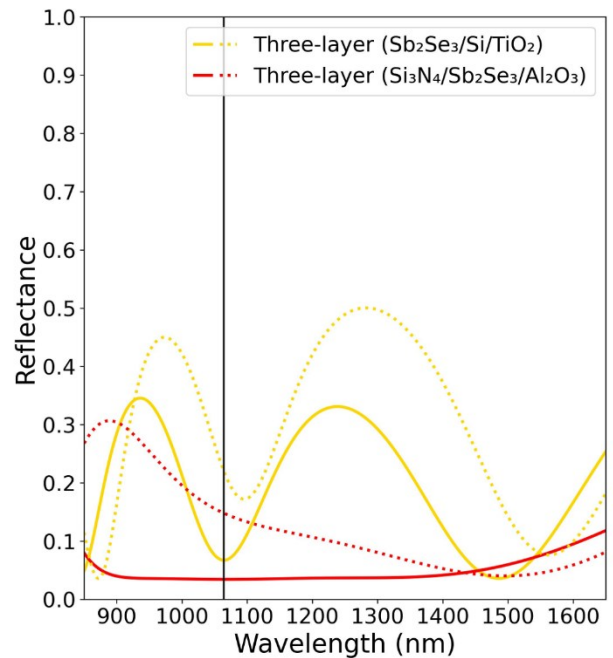
So, we need a new structure that has high reflectance contrast but still maintain low reflection in amorphous phase. This introduces us to edge-filter design, the reflectance profile has HR-region centered around the reference wavelength, while the regions outside of this are both AR regions, with a steep spectral edges between the AR- and HR-regions. Edge filter consists of QWOT-thick high-index film (H) and low-index film (L) deposited on top of each other in a structure called unit cell (U), it can either be a long-pass ($H/2 \ L \ H/2$) or a short-pass designs ($L/2 \ H \ L/2$) the $1/2$ symbol denotes half QWOT thickness ($OT = \lambda/8$). The spectral width of HR-region is directly correlated to the difference between H and L layer, and the number of unit cell directly correlated to the steepness of spectral edge between region. The long-pass design has stable, low reflectance in the longer-wavelength AR region, while short-pass design has the opposite behavior. The long-pass design is selected for this work since it minimize the device thickness. And since we want our devices to operate around 1064 nm, the reference wavelength is moved to around $\lambda_0 = 740$ nm. This way, the interested wavelength ($\lambda = 1064$ nm) is at the edge of AR-region, so when Sb_2Se_3 transitioned to crystalline phase, the reflection profile red-shift and the interested wavelength is moved to high-reflectance region.

As an examples, Sb_2Se_3 is H layer, and MgF_2 as L layer, the reference wavelength is changed to $\lambda_0 = 740$ nm. Figure 5 illustrates the effect the number of unit cell has on the reflectance profile. By increasing the number of unit cell from 1 to 6, the spectral edge is getting steeper. Also, the interested wavelength (black vertical-line) shifted from AR-region in amorphous phase (Fig. 5a) to HR-region in crystalline phase

(Fig 5b), this create a reflectance contrast higher than single-, double- or three-layer designs previously shown.

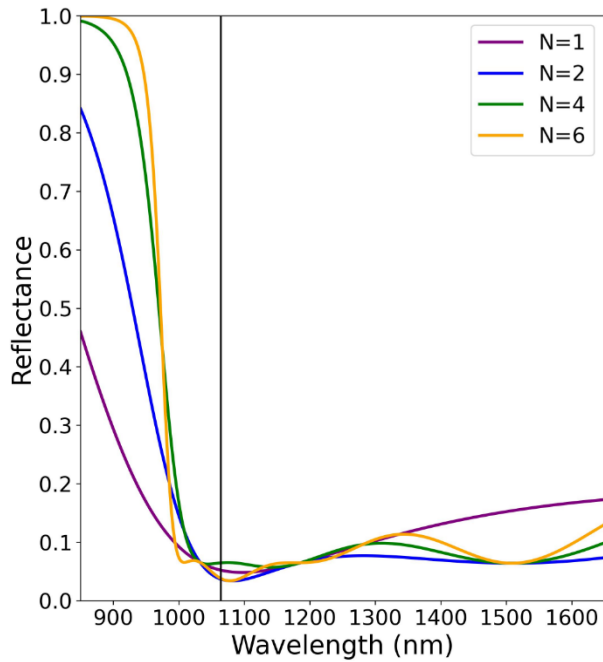


(a) Reflectance of single and two-layer designs

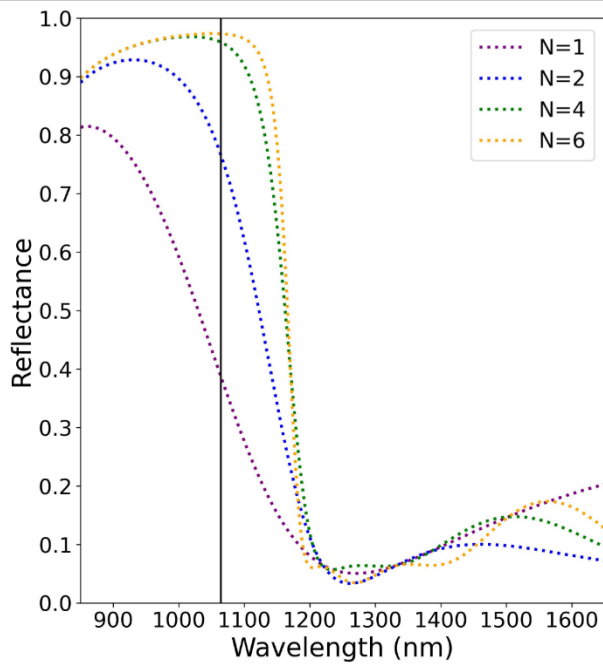


(b) Reflectance of different three-layer designs

FIGURE 4. Reflectance of the designs in Fig. 3 before (solid) and after (dotted) phase transition. The single-layer design has the highest reflectance contrast ($\Delta R = R_a - R_c$) of 55.0% .



(a) Amorphous phase



(b) Crystalline phase

of each design. The computation codes are based on [26], but other simulation software are also applicable. The simulations are going to be from 850–1650 nm at an interval of every 1 nm. The dataset of most element in this work can be found in the website *refractiveindex.info* [30], with the only exception being Sb_2Se_3 [19]. The high-index layers (H) are silicon with refractive index at 1064 nm equal to $n_{\text{Si}} = 3.56$ [28], titanium dioxide ($n_{\text{TiO}_2} = 2.31$) [27], and silicon nitride ($n_{\text{Si}_3\text{N}_4} = 2.00$) [29], while the low-index layers (L) are aluminium oxide ($n_{\text{Al}_2\text{O}_3} = 1.62$) [27], silicon dioxide ($n_{\text{SiO}_2} = 1.47$) [25], and magnesium fluoride ($n_{\text{MgF}_2} = 1.38$) [26]. The substrate layer has the same refractive index as n_{SiO_2} and is 3 mm thick, reflection from the substrate-air layer on the backside is calculated. Every dataset used here has a measuring points from 350–1650 nm at an interval of ≤ 4 nm. And the incident angle is 0° , and light polarization mode is transverse electric (TE), unless specifically mentioned.

The main performance indicators are going to be transmittance at 1064 nm in both phases and the transmittance contrast ($\Delta T = T_{\text{ON}} - T_{\text{OFF}}$). The secondary performance indicators are; the high-contrast region spectral width, the performance at different polarization and incident angle, the total PCM used, design thickness, and the complexity of the design which is indicated by the number of layers and the number of material types used.

B. SIMULATION OF OPTICAL ENDURANCE AND STABILITY UNDER LARGE PHASE SWITCHING CYCLES

Experimental studies have demonstrated that Sb_2Se_3 can has phase switching cycles with minimal degradation of its optical contrast and maintain a low-loss, reversible modulation in the order of 10^6 phase switching cycles [23]. Even so, Sb_2Se_3 still exhibits a slight complex refractive index drift due to structural relaxation and cumulative thermal damage. Accounting for these variations is essential for ensuring long-term reliability and precision in high-endurance photonic applications [23, 24]. To account for this, the complex refractive index ($\tilde{n}$) can be rewritten as

$$\tilde{n}_p(C) = n_p(C) + ik_p(C), \quad (17)$$

where C is the number of the phase switching cycle. The real part of the refractive index (n) is described by an exponential relaxation towards a saturated value such that

$$n_p(C) = n_{p,\infty} - (n_{p,\infty} - n_{p,0})e^{-\alpha_p C}, \quad (18)$$

where $n_{p,\infty}$ is the asymptotic refractive index after many cycles, $n_{p,0}$ is the refractive index at the first cycle (obtained from ellipsometry or literature data), and α_p is the drift coefficient per cycle of phase p [21, 35].

Meanwhile, the imaginary part k can be approximated from a linear growth as

FIGURE 5. Reflectance profile of long-pass edge-filter designs in each phases. The design used increasing number of unit cell, from 1–6 unit cells. (a) When devices are in amorphous phase (ON mode), the wavelength of interest (black vertical line) is in the AR region. (b) When devices are changed to crystalline phase (OFF mode), the wavelength of interest is in the high-reflection region.

III. METHODS

A. SIMULATION OF OPTICAL PERFORMANCES

We use transfer matrix method (TMM) of Fresnel equation to calculate the transmittance, absorptance, and reflectance

$$k_p(C) = k_{p,0}(1 + \beta_p C), \quad (19)$$

where $k_{p,0}$ is the initial loss, and β_p is the loss growth coefficient. Parameter β_p is the relative loss per cycle of phase p [23, 36]. The model from Eq. (19) is adopted in this work, they are consistent with experimental trends and with general PCM endurance theory, providing a practical way to assess contrast degradation and threshold drift of PCM-based devices over large cycles [21, 23, 35, 36].

Optical endurance can be quantified using the following metrics; percentage transmitting contrast degradation (D_{pct}) and insertion loss (IL). Transmittance contrast degradation is defined as the change in transmittance contrast after N cycles:

$$D_{pct}(C) = \frac{\Delta T(C) - \Delta T_0}{\Delta T_0}, \quad (20)$$

where ΔT_0 is the transmittance contrast between the amorphous and crystalline phases in the initial cycle $C = 0$ ($\Delta T_0 = T_a(0) - T_c(0)$). A negative D_{pct} indicates a decreasing contrast.

The insertion loss (IL) of the device at cycle C compares the high-transmittance amorphous phase after the initial cycle:

$$IL_{dB}(C) = 10 \log \left[\frac{T_a(C)}{T_a(0)} \right]. \quad (21)$$

A increasing $IL_{dB}(C)$ indicates increasing loss, which is typically driven by the linear growth of the imaginary index $k_p(C)$ and by detuning of the resonance due to drift in $n_p(C)$.

IV. RESULTS & DISCUSSIONS

A. OPTICAL PROPERTIES OF 1064-NM OPTICAL FILTERS

The design architectures, number of layers, total thickness of Sb_2Se_3 , overall design thickness, transmittance in each

mode, and transmittance contrast ($\Delta T = T_a - T_c$) of each design are shown in Table I. The operating wavelength is 1064 nm, the reference wavelength (λ_0) is 1064 nm for single-, two-, and three-layer designs, but $\lambda_0 = 740$ nm for all edge-filter designs. The simulation show that only edge-filter designs with at least three unit-cells ($N \geq 3$) have ΔT higher than 50%, with 6-unit-cell design having the highest ΔT of 94.5%. However, while there is only marginal improvement in ΔT when N exceeds 3, the Sb_2Se_3 layer thickness and the overall device thickness increase drastically; for example, ΔT improves by 2% in $N = 4$ design when compared to $N = 3$ design, the thickness increases by 33.3%, but when comparing $N = 5$ to $N = 4$ designs, ΔT increases by only 0.6%, and the device needs to be 25.0% thicker. It is important to note that increasing unit cell numbers also require more Sb_2Se_3 material, less Sb_2Se_3 used also has the benefit of minimizing the energy required for phase transition.

In edge-filter designs, the transmittance in OFF mode decrease with increasing N until it reaches zero in 6-unit-cell design. This is because, unlike the amorphous phase, crystalline phase Sb_2Se_3 still has a small absorption at 1064 nm ($k_{Sb_2Se, cry} = 0.0097$), so light is absorbed by each layer until T_{OFF} is reduced to zero. For context, the absorption of crystalline Sb_2Se_3 is still an order of 10^2 smaller than other PCMs, e.g. GST ($k_{GST, cry} = 2.4540$) or VO2 ($k_{VO2, metallic} = 1.4580$). Figure 4.1 shows the transmittance contrast of the single-layer design and edge-filter designs ($1 \leq N \leq 6$). Even though the ΔT at 1064 nm edge filter designs with $N \geq 2$ are around the same value, the spectral width of high-contrast region is noticeably wider in higher unit-cell designs. This property should be considered if the filter is intended to be used in a broad spectrum applications.

TABLE I
DESIGNS OF Sb_2Se_3 -BASED OPTICAL FILTERS

Design	Layout	Number of layer	Sb_2Se used (nm)	Device size (nm)	T_{ON} (%)	T_{OFF} (%)	ΔT (%)
Single-layer	Sb_2Se_3	1	157.9	157.9	93.5	37.2	56.3
Two-layers	Sb_2Se_3/TiO_2	2	78.9	194.1	93.3	76.9	16.4
Three-layers	$Sb_2Se_3/MgF_2/TiO_2$	3	78.9	580.3	93.3	76.9	16.4
Three-layers	$Si_3N_4/Sb_2Se_3/Al_2O_3$	3	157.9	455.7	96.6	83.3	13.3
Edge filter	$(\frac{Sb_2Se_3}{2}/MgF_2/\frac{Sb_2Se_3}{2})^1$	3	48.5	182.5	94.1	54.3	39.8
Edge filter	$(\frac{Sb_2Se_3}{2}/MgF_2/\frac{Sb_2Se_3}{2})^2$	5	97.0	365.0	95.0	16.3	78.7
Edge filter	$(\frac{Sb_2Se_3}{2}/MgF_2/\frac{Sb_2Se_3}{2})^3$	7	145.5	547.6	94.8	4.0	90.8
Edge filter	$(\frac{Sb_2Se_3}{2}/MgF_2/\frac{Sb_2Se_3}{2})^4$	9	194.0	730.1	93.7	0.9	92.8
Edge filter	$(\frac{Sb_2Se_3}{2}/MgF_2/\frac{Sb_2Se_3}{2})^5$	11	242.6	912.6	93.6	0.2	93.4
Edge filter	$(\frac{Sb_2Se_3}{2}/MgF_2/\frac{Sb_2Se_3}{2})^6$	13	291.1	1,095.1	94.5	0.0	94.5

Transmittance is calculated at $\lambda = 1064$ nm. Single, two, and three-layer designs all have $\lambda_0 = 1064$ nm, but all edge-filter designs have $\lambda_0 = 740$ nm

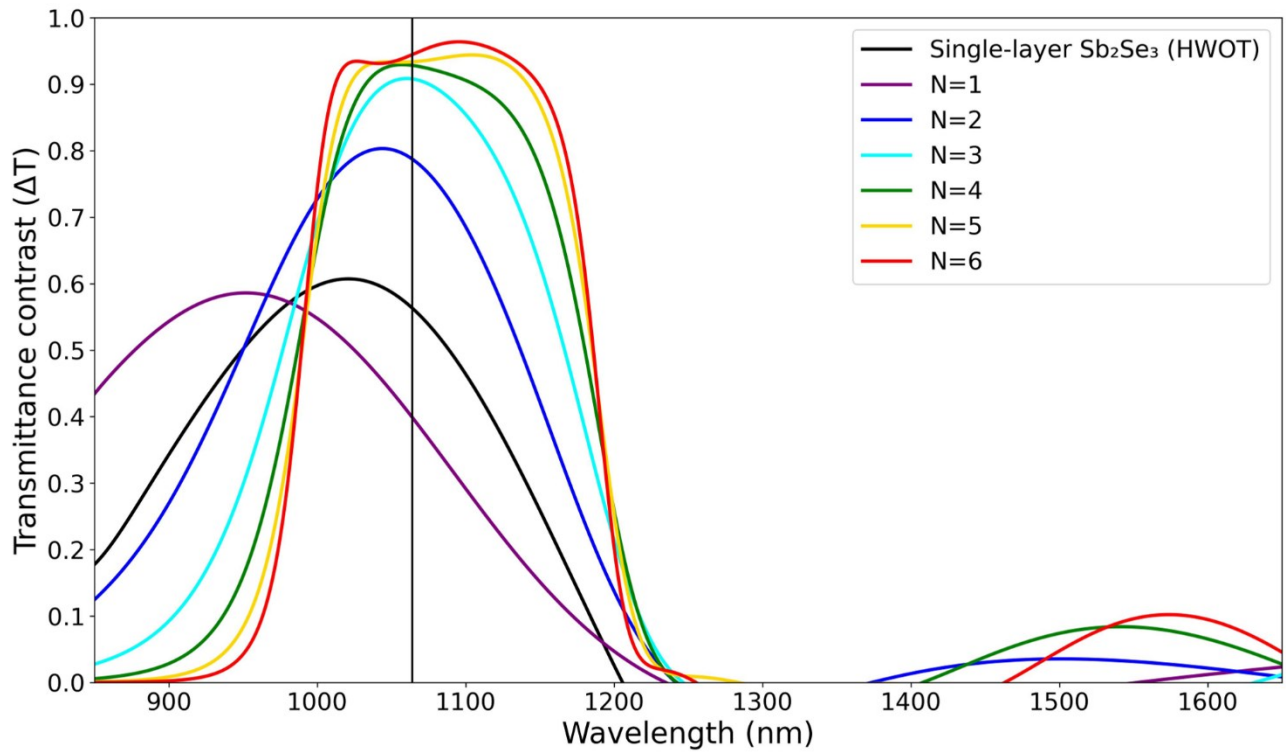


FIGURE 6. Transmittance contrast (ΔT) of a single-layer and edge-filter designs with different number of unit cells (N). The vertical black line indicates the interested 1064 nm wavelength.

Two other properties that might need to be considered are the optical performance at varying incident angles and the polarization mode of the incident light (TE or TM). Figure 7 shows 4-unit-cell and 5-unit-cell designs with incident angles ranging from 0° to 90° , and with polarization in TE (Fig. 7a and 7c) and TM (Fig. 7b and 7d). For both TE and TM polarized light, ΔT remains higher than 90% from 0° to $\sim 30^\circ$, but TM polarization causes the high- ΔT region to blueshift significantly at higher incident angle. The spectral widths of the high- ΔT region in both designs are comparable, but the 5-unit-cell design has a higher ΔT across the spectrum, which indicates a better broadband, wide-angle, tunable optical device.

With the reasons given above, we decided that a 5-unit-cell, long-pass edge-filter design (Fig. 6, yellow) has the best compromise between high optical performances, device size, and complexity. This design has a broadband transmittance contrast between the amorphous (ON mode) and the crystalline phases (OFF mode), and the FWHM of high- ΔT region is 197.0 nm. This design is used as an example in the following sections.

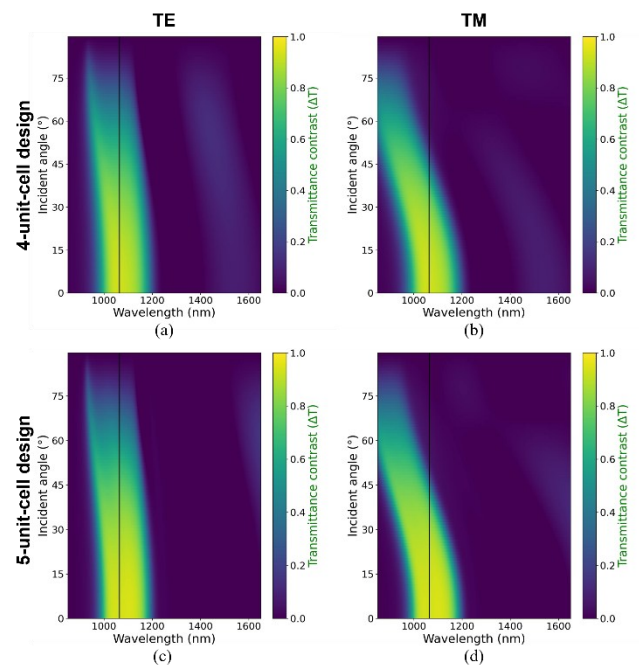


FIGURE 7. Transmittance contrast at different incident angles and wavelengths of 4-unit-cell and 5-unit-cell long-pass edge-filter designs.

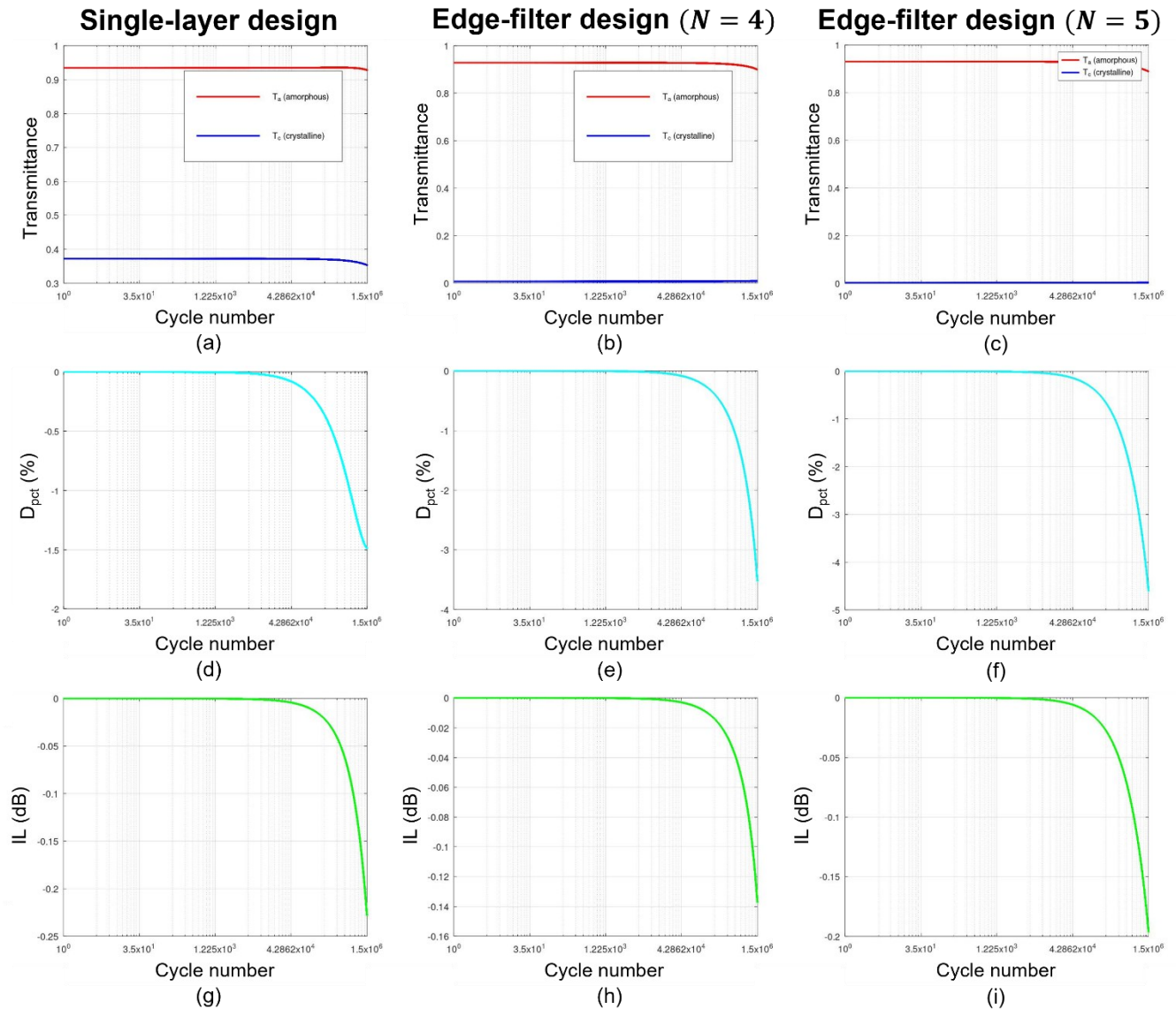


FIGURE 8. Optical endurance at 1064 nm of the single-layer, and 5-unit-cell edge-filter designs after 1.5×10^6 phase-switching cycles. (a,b) Transmittance in amorphous (red) and crystalline phases (blue). (c,d) The percentage transmittance contrast degradation relative to the initial value. (e,f) Insertion loss in amorphous phase, expressed in dB. Note a different y-axis scales.

B. Endurance simulation of 1064-nm optical filters(check endurance of edge-filter.)

Endurance simulation at 1064 nm of the single-layer Sb_2Se_3 , and 5-unit-cell edge-filter designs demonstrated that all three resonant optical filters remain operational up to 1.5×10^6 phase switching cycles, with design-dependent degradation of their transmittance metrics. For the single-layer design, both the amorphous (T_a) and crystalline phases transmittance (T_c) decrease modestly (Fig. 8a). And both the corresponding percentage transmittance contrast degradation (D_{pct}) and insertion loss (IL) also decrease with cycling number, as shown in Fig. 8d and 8g, respectively.

For both the 4- and 5-unit-cell edge-filter designs, the T_a and insertion loss decrease at a rate slower than in the single layer design, and their T_c increased slightly near the end. This causes

their percentage transmittance contrast degradation to become two and three times larger, Fig. 8e and 8f, respectively. Their larger D_{pct} are attributed to the fact that their T_c become higher at the latter cycle. Edge-filter designs have a stronger field confinement so they exhibit less insertion losses when compared to the single-layer design.

These simulation results confirm that designs with multiple layers of Sb_2Se_3 still maintains comparable transmittance endurance to a simple, single-layer design. Their larger D_{pct} might become a problem at larger switching cycle but is overall negligible in the confine of our simulation. Overall, the edge-filter designs still have a higher transmittance contrast and a lower insertion loss after 1.5×10^6 cycles.

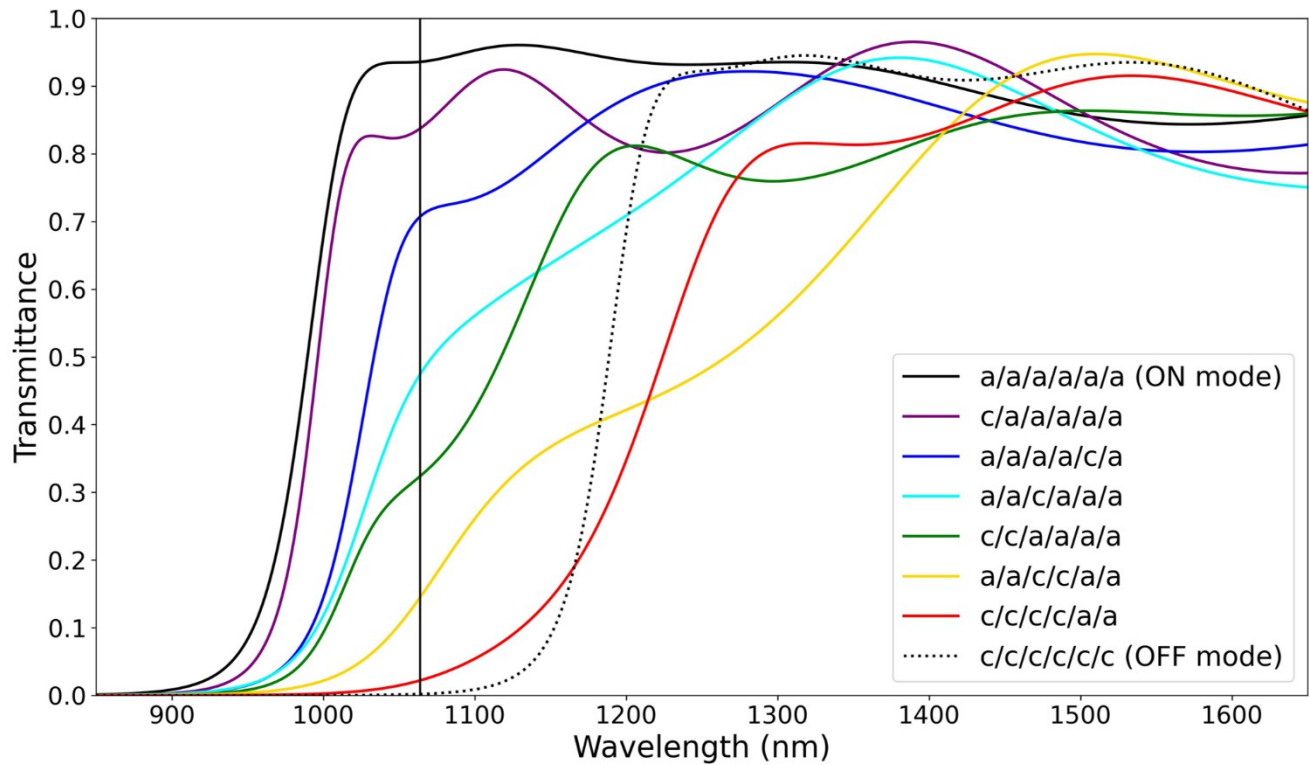


FIGURE 9. Transmittance of the five-unit-cell edge-filter design in 8 different configurations. Each Sb_2Se_3 layer is in amorphous (a) or crystalline (c) phases, and the layout ordering from left to right is substrate-side to air-side.

4.3 Transmittance tunability and optimization.

Edge-filter designs have multiple layers of Sb_2Se_3 which complicates the phase switching setup. When laser-induced heating is used as a switching method, the phase transition of every Sb_2Se_3 layer does not occur simultaneously and the transmission changes gradually. This effect is shown in the optical limiter of Sarangan, et al. [3] Their device use a similar long-pass edge-filter design but the H layers are GST instead of Sb_2Se_3 , and the L layers are SiO_2 instead of MgF_2 . Their experimental results showed that the phase switching of GST layers occurs layer-by-layer in an order that is determined by the changes in the optical field distribution and thermal crosstalk between layers, which does not necessarily start from layer in the air -side to the substrate-side (top to bottom). However, our device is intended to be used as a transmittance-tunable filter that does not require a fast switching speed; therefore, electrical heating is also available as an alternative phase switching method. The ability to manually control the phase of each Sb_2Se_3 layer enables the filter to have multiple transmittance values. Using the 5-unit-cell edge-filter design described in Table I, which consists of 11 layers of Sb_2Se_3 and MgF_2 (six layers of Sb_2Se_3), there would be a total of 2^6 configurations for the Sb_2Se_3 phases. Figure 9 highlights some of these configurations; the legend denotes Sb_2Se_3 in the amorphous (a) or crystalline (c), and the ordering from left to right is from the substrate to the

air side (f_1 to f_6). With the correct phase configuration, the transmittance can be gradually reduced from peak of 93.4% to 0.0%.

The peak transmittance contrast of each design shown in Fig. 6 does not always correspond to the desired operating wavelength of 1064 nm. This is explained by the arbitrary selection of the reference wavelength in edge-filter designs. So, an optimization algorithm can be used to improve the optical performance. A simple merit function found in OpenFilters [33] is used in this example. The optimization goal for the 5-unit-cell edge-filter design is to have a transmittance value of at least 97% from 1000 to 1200 nm at intervals of every 20 nm. The optimization took less than 5 s, and the new design now required 304.2 nm of Sb_2Se_3 , 25% more than the original design, but the overall thickness is now 845.6 nm, 7% less than the original, the thickness of each layer is shown in Table II. Figure 10a shows the transmittance of the original (green) and optimized designs (blue) in the amorphous (solid) and crystalline phases (dotted). Overall, the optimized design has a higher transmittance in the desired 1000 to 1200 nm spectrum, but the spectral profile has been blue shifted by a small amount. The transmittance contrast at 1064 nm (Fig. 10b) of the optimized design is also higher, 95.8%, compared to 93.4% of the original. The transmittance contrast FWHM also extended slightly to 207.4 nm (original: 197.0 nm)

TABLE II
THICKNESS OF EACH LAYER BEFORE AND AFTER OPTIMIZATION

Design	SiO ₂	Sb ₂ Se ₃	MgF ₂	Sb ₂ Se ₃	MgF ₂	Sb ₂ Se ₃	MgF ₂	Sb ₂ Se ₃	MgF ₂	Sb ₂ Se ₃	MgF ₂	Sb ₂ Se ₃
Original	3 mm	24.3	134.0	48.5	134.0	48.5	134.0	48.5	134.0	48.5	134.0	24.3
Optimized	3 mm	46.1	67.5	59.3	148.1	55.0	89.3	50.0	145.3	63.3	91.2	30.5

Original design is the 5-unit-cell long-pass edge filter. The units of Sb₂Se₃ and MgF₂ thickness are nm.

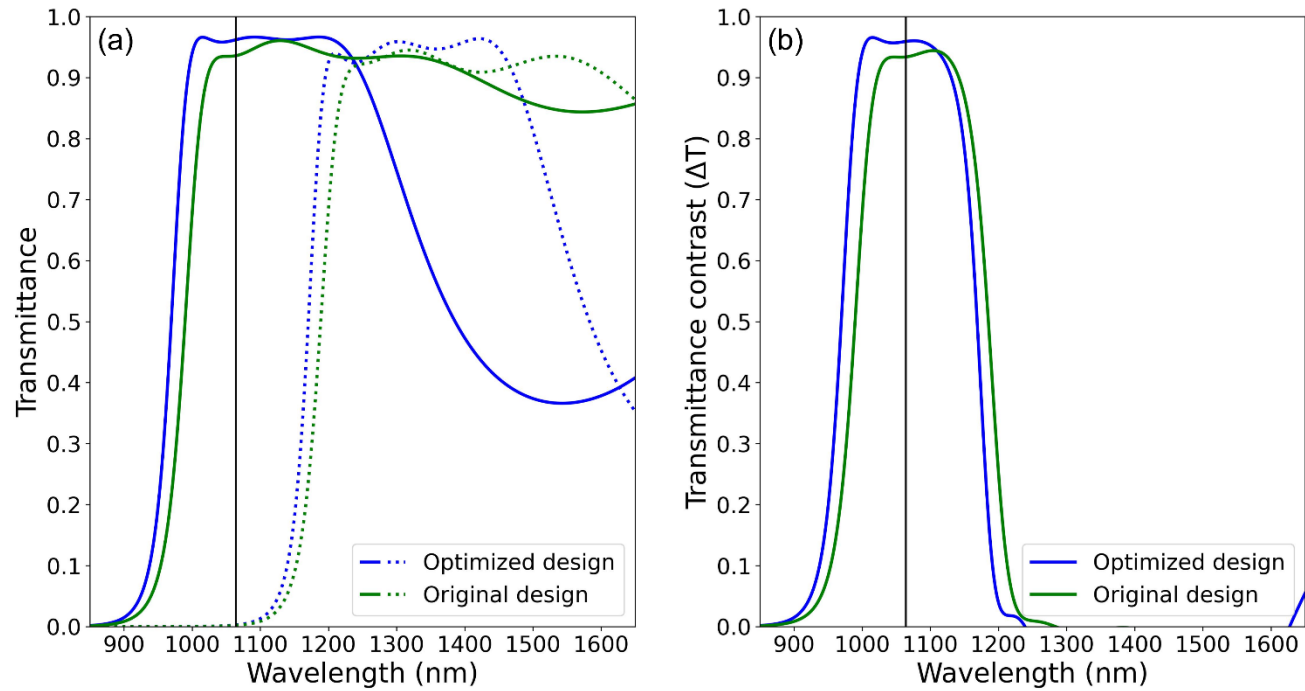


FIGURE 10. Transmittance comparison of 5-unit-cell edge-filter design before (green) and after (blue) optimization. (a) The transmittance in ON mode (solid) and OFF mode (dotted). (b) The transmittance contrast (ΔT). An optimized design has higher optical performance from 900 to 1200 nm, but the high- ΔT region has been blueshifted by a small amount.

V. Conclusions

We have shown that Sb₂Se₃ is a suitable PCM for high-efficient, broadband transmittance-tunable optical filter operating around 1064 nm wavelength. The 5-unit-cell long-pass edge-filter design has the best compromise between optical performance, size, and design complexity. This design required six Sb₂Se₃ layers and five MgF₂ layers and is 912 nm thick. Transmittance at 1064 nm in the ON mode is 93.4% and 0.2% in OFF mode, a contrast of 93.4% with around 200 nm FWHM. Simulations show that after 1.5×10^6 phase switching cycles, the transmittance contrast degradation and insertion loss in ON mode is still within acceptable ranges. The multilayer nature of edge-filter design enables the gradual modulation of transmittance by changing the phase of each Sb₂Se₃ layer. We also show that a quick and simple optimization algorithm can increase the optical performance from 900-1200 nm.

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DISCLOSURE

The authors declare no competing interests.

DATA AVAILABILITY

Data underlying the results presented in this paper may be obtained from the corresponding author upon a reasonable request.

DECLARATION OF AI USAGE

Gemini was used for code debugging and grammar correction. Consensus was used to help with the initial literature review. The authors have reviewed and edited the content as needed after using these tools.

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